MECHANISTIC ANALYSIS

Contents

- Coupler Curves
- Toggle Position
- Transmission Angle
- Mechanical Advantage
- Acceleration Analysis
- Coriolis Component of acceleration

Coupler Curves

- Def:
  - A coupler curve is the locus of a point on the coupler link.
- For the mechanisms considered, the displacement of the links joined with the fixed link was the input or output of the simple mechanisms.

Coupler Curves

- In great number of applications the output from a simple mechanism is the path traced by one of the points on the coupler link.
- These paths are generally called "***coupler point curves***" or "***coupler paths***".

$$\delta = \frac{180° - \gamma}{2}$$

$$AP = 2(AB)\cos\delta$$

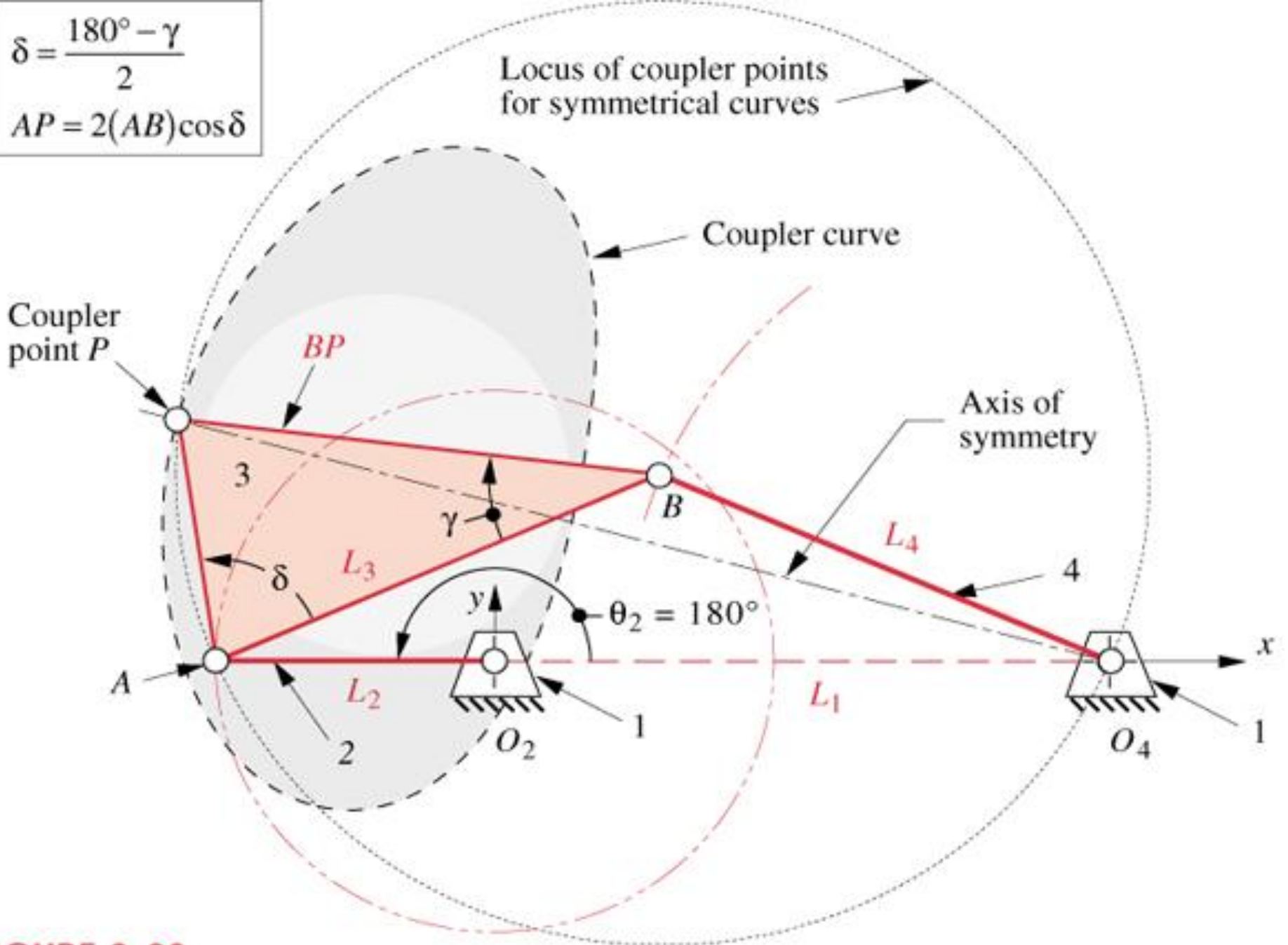
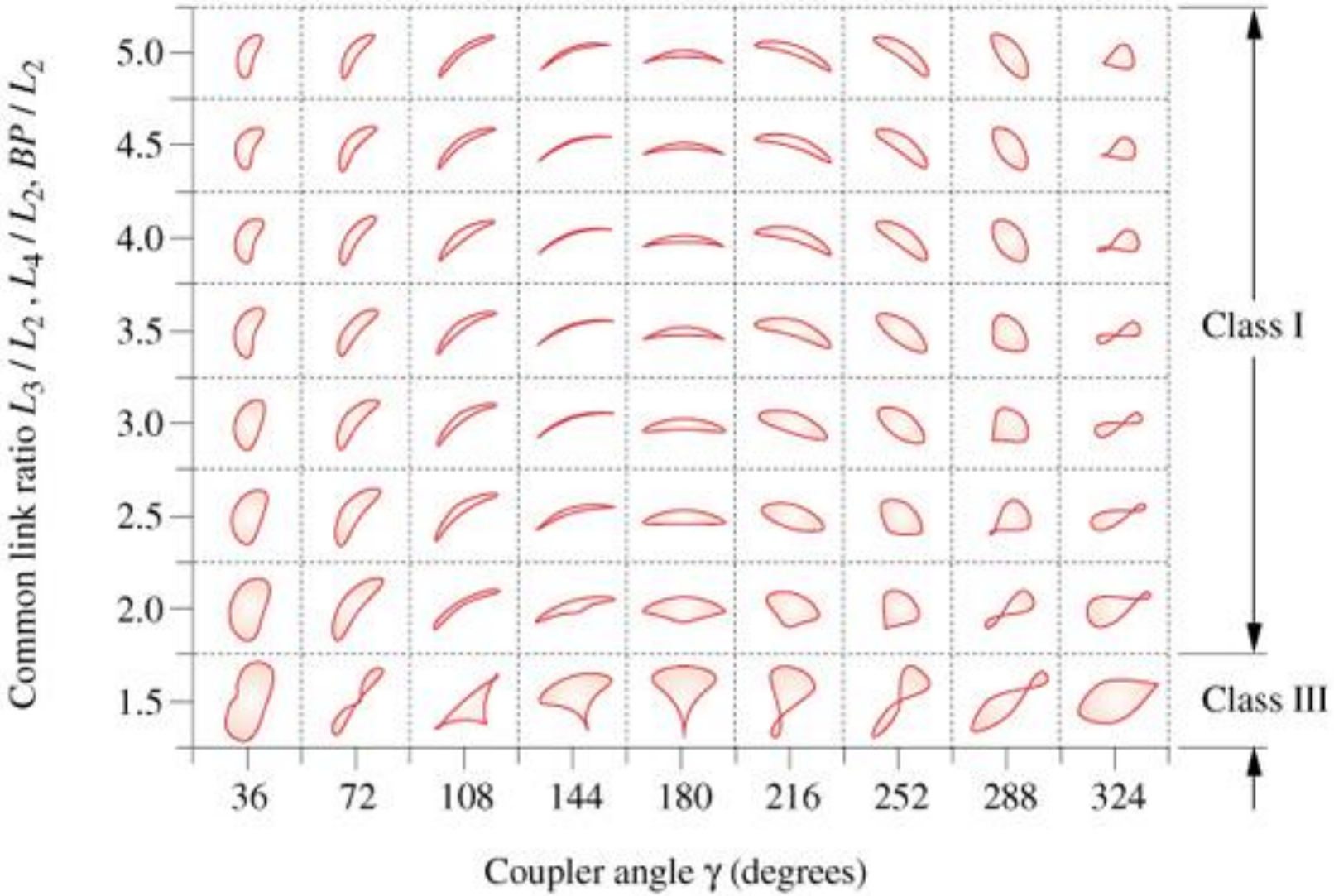


FIGURE 3-20

A fourbar linkage with a symmetrical coupler curve.



(*a*) Variation of coupler curve shape with common link ratio and coupler angle for a ground link ratio $L_1 / L_2 = 2.0$

FIGURE 3-21

Coupler curve shapes of symmetrical fourbar linkages *Adapted from reference [9]*

Toggle Position

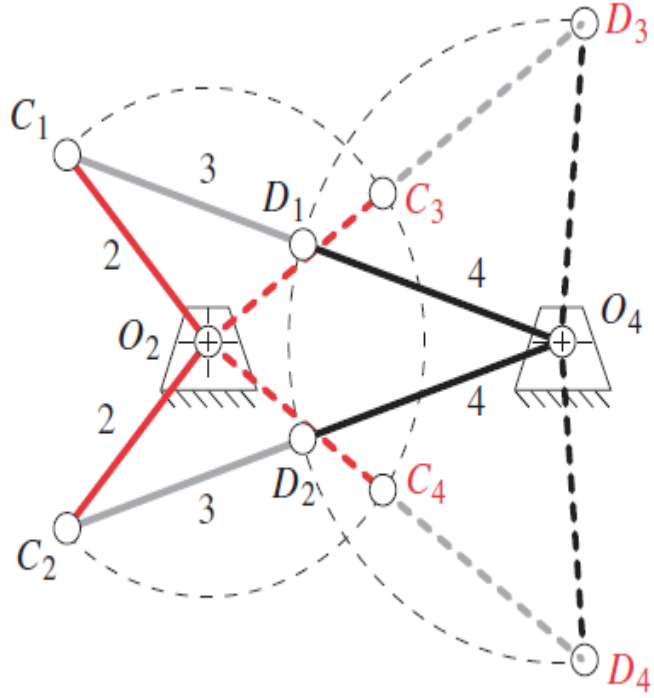
- There are various, methods and synthesis that could be used to obtain trail solution to motion control problem.
- One of such rapid process is toggle position.
- In this technique, one needs to check whether the mechanism reaches all the desired position without encountering a limit position.

Toggle Position

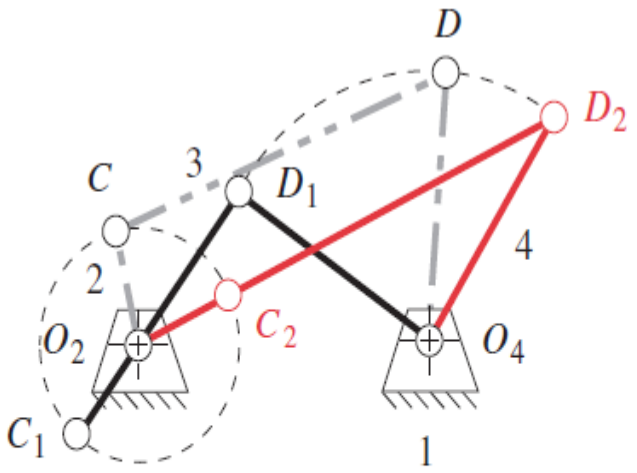
- The toggle positions are determined by the co-linearity of two of the moving links.
- Hence any mechanism at its limit of motion is known as Toggle position.

Toggle Position

- Example showing toggle position technique.



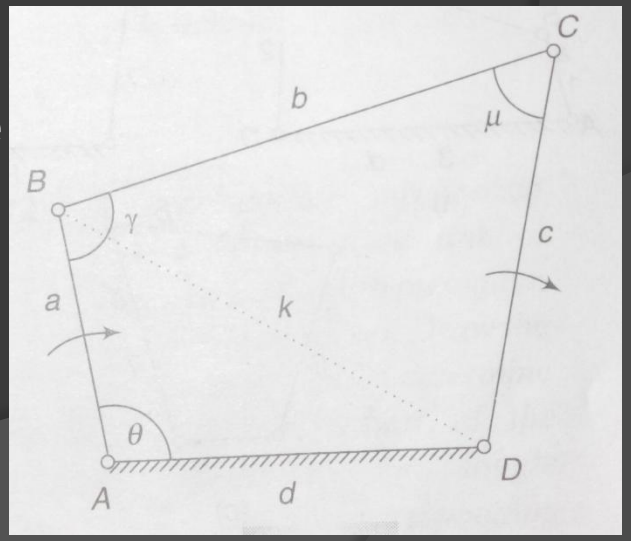
(*a*) Non-Grashof triple-rocker toggle positions



(*b*) Grashof crank-rocker stationary configurations

Transmission Angle

- The angle μ between the output link and the coupler is known as transmission angle.
- As shown in fig, if link AB is input & the force applied to output DC is transmitted through coupler BC.
- For particular value of force, the torque is max when μ is 90º.

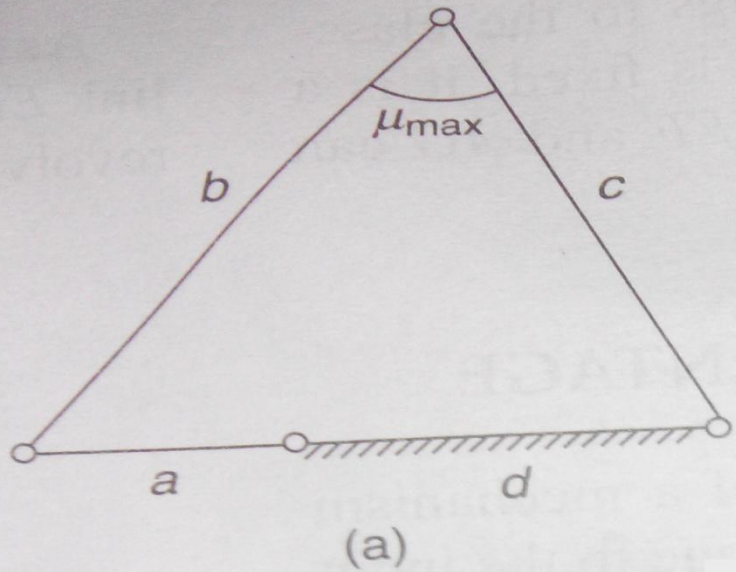


Transmission Angle

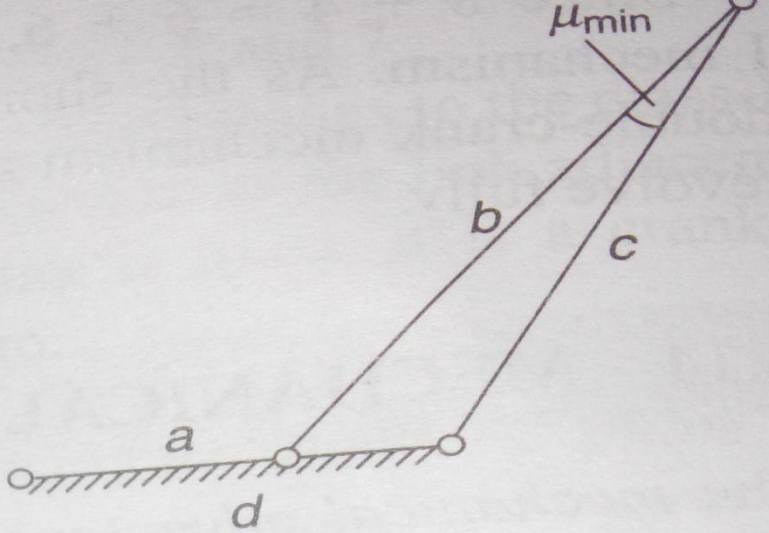
- When BC and DC are coincident μ is 0 and the mechanism would lock.
- When μ deviates a lot from 90º, the torque on output link decreases.
- Generally μ is kept more than 45º.
- The max and min value of μ can be found by putting dμ/dθ equal to 0.

Transmission Angle

- It can be seen that μ is max when θ is 180º, & min when θ is 0º.
- But this happens only in double crank or rocker crank mechanism.
- The max and min value of μ is both mechanisms are given in next slide.

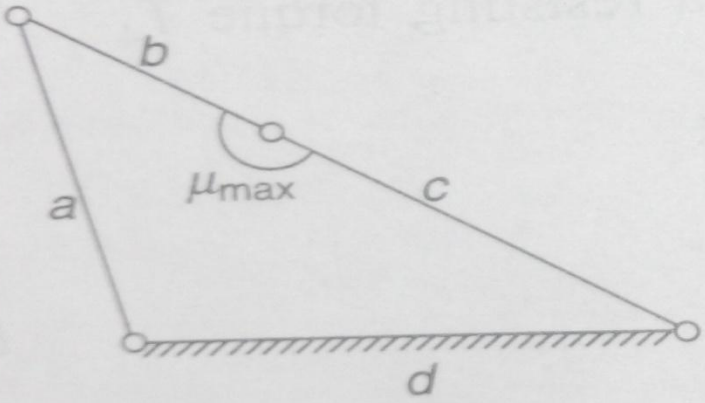


(a)

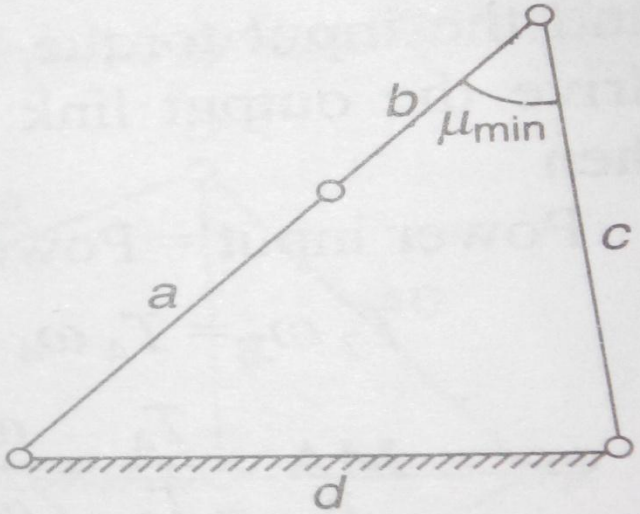


(b)

Crank-Rocker Mechanism



(a)

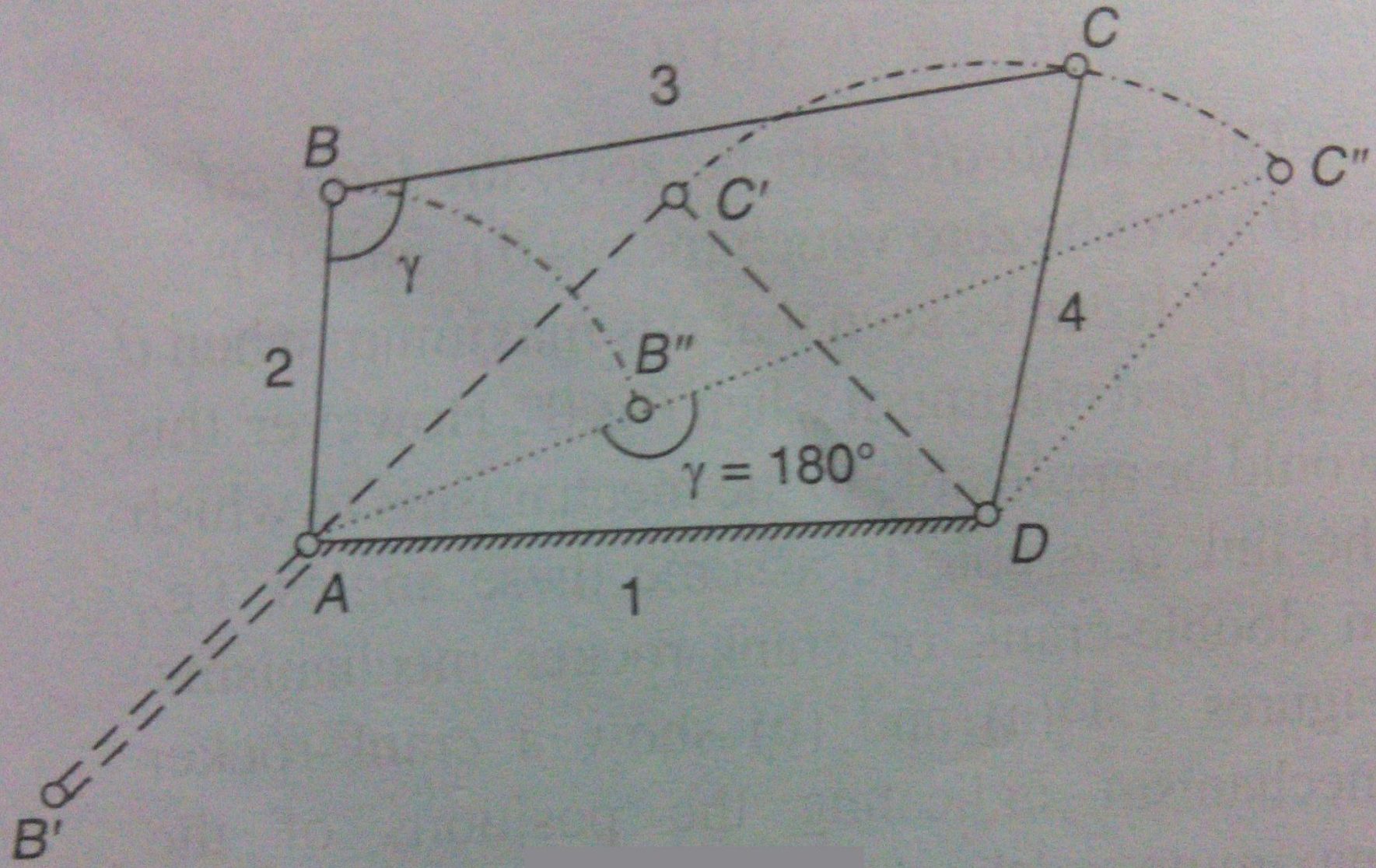


(b)

Double Rocker Mechanism

Mechanical Advantage

- The mechanical advantage (MA) of a mechanism is the ratio of the output force/torque to the input force/torque at any instant.
- In the given fig, is friction and inertia forces are ignored,
  - Input torque T_2 is applied to link 2
  - Output resisting torque T_4 produced at link 4


C
3
B
C''
C'
γ
4
2
B''
γ = 180°
D
A
1
B'

Mechanical Advantage

- Power input = Power Output

$$T_2\ \omega_2 = T_4\ \omega_4$$

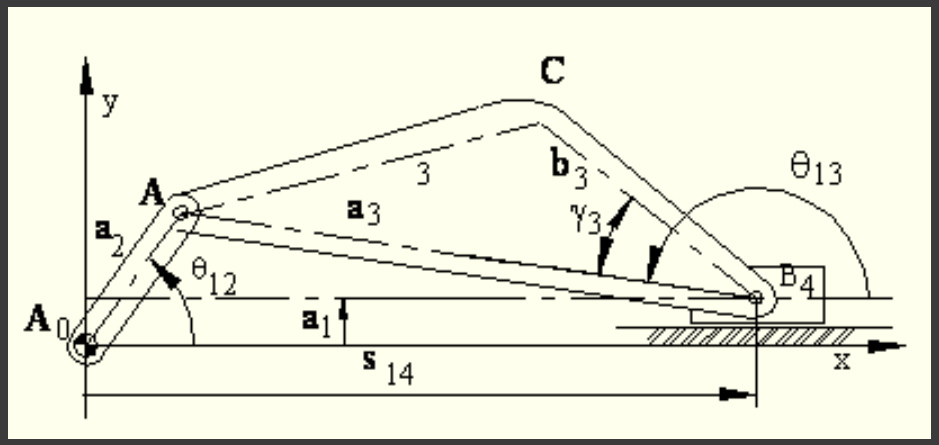
- Or MA = $T_2 / T_4 = \omega_2 / \omega_4$
- Thus it is reciprocal of velocity ratio.
- But in case of crank-rocker, ω_4 is 0, and γ becomes 180º or 0º, making MA infinity.
- Hence only a small input torque can overcome a large output torque load.
- Extreme position of linkage points are known as Toggle points.

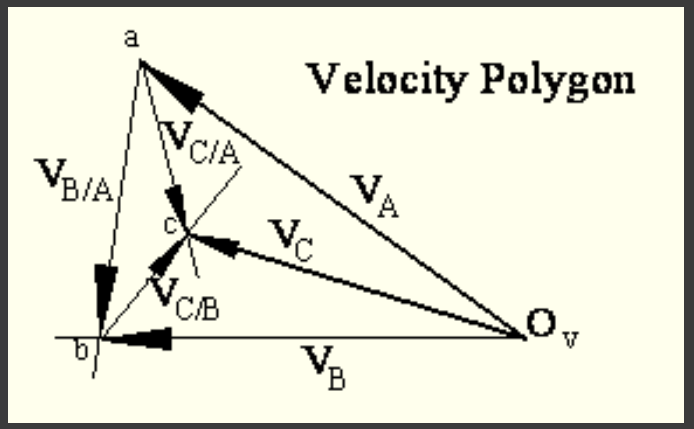
Acceleration Analysis

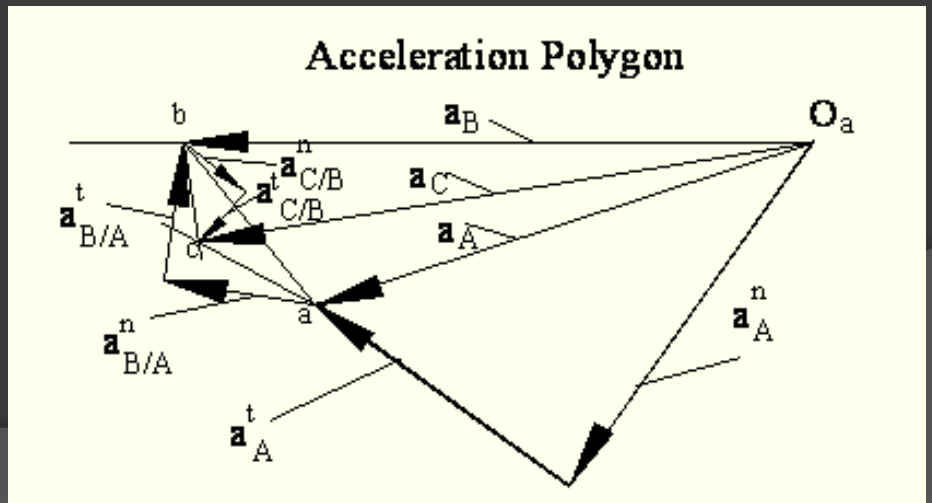
- Acceleration analysis of mechanisms can be performed vectorially using the relative acceleration concept, usually starting with the given values and work through the mechanism by way of series of points A, B, C, etc.
- important consideration is that the acceleration analysis cannot be performed without performing the velocity analysis since the normal and Coriolis acceleration components can only be determined after the velocity analysis.

Acceleration Analysis

- An Example.





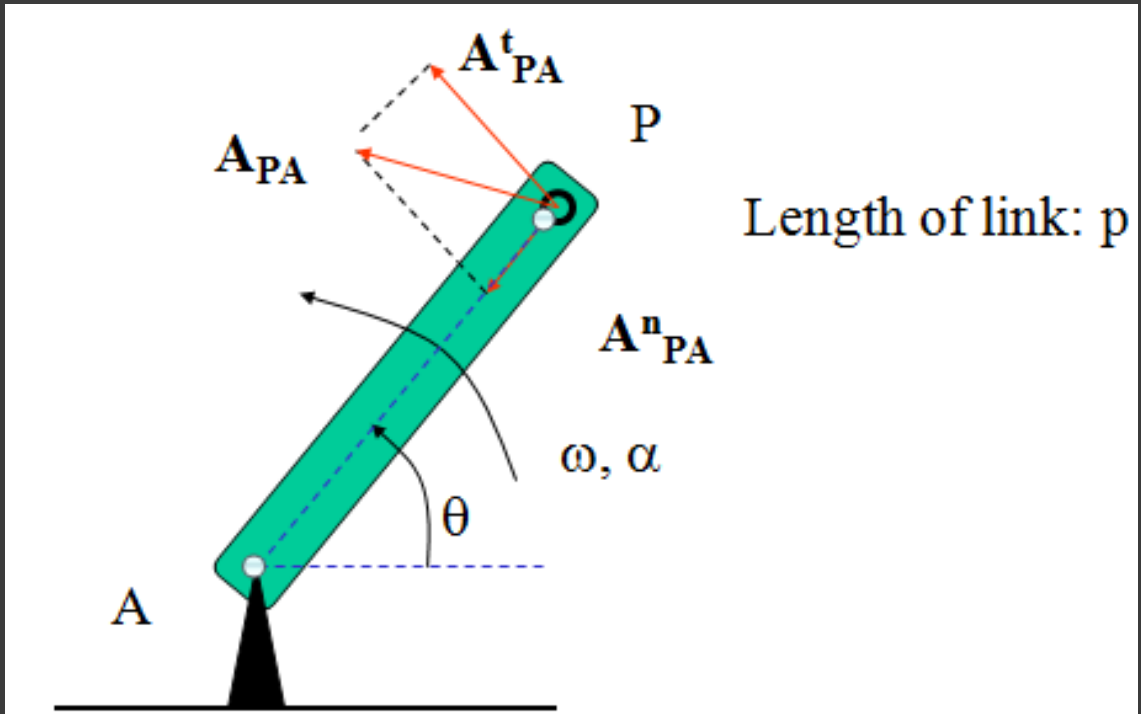


Acceleration Analysis

- The triangles abc formed on the velocity and acceleration polygons are similar to the triangle ABC of the mechanism link. The sense of abc is similar to the sense of ABC

Acceleration Analysis

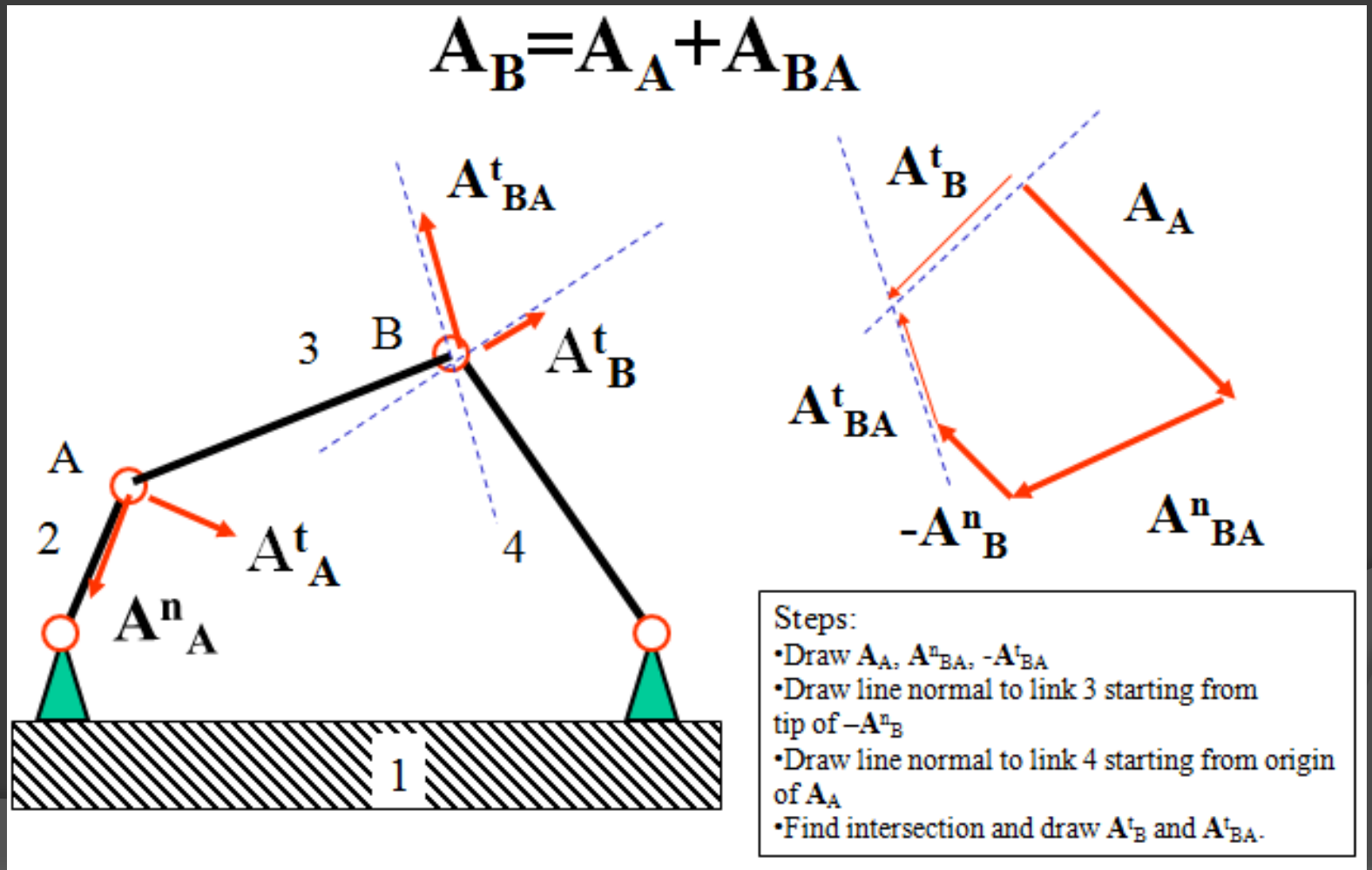
- Acceleration of a link in pure rotation.



Magnitude of tangential component = pα,
magnitude of normal component = p ω^2

Acceleration Analysis

- An Example.



Coriolis Component of Acc.

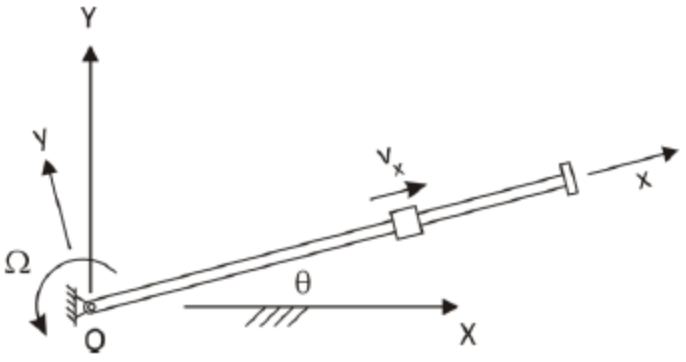
- By derivation of the five-term acceleration equation with a spinning and translating reference frame, we see that there are two terms that appear when an object moves inside this rotating frame. One of these is Coriolis acceleration.
- The Coriolis term is

$$\vec{a}_{Cor} = 2\vec{\Omega} \times \vec{v}_{xy}$$

- Components of Coriolis acceleration involve only two velocities
  - The rotational velocity of the rotating frame,
  - the velocity of an object within this rotating frame

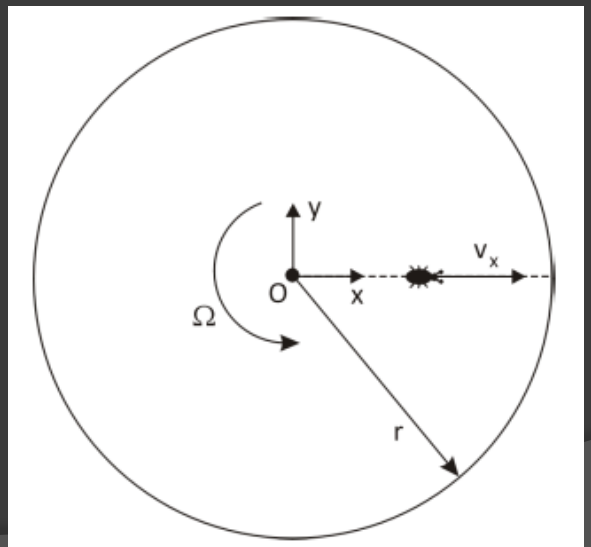
Coriolis Component of Acc.

- Example a rotating rod with a collar moving along the rod.
- To understand coriolis component of acc. Let us take an example.



Rotating rod with sliding collar

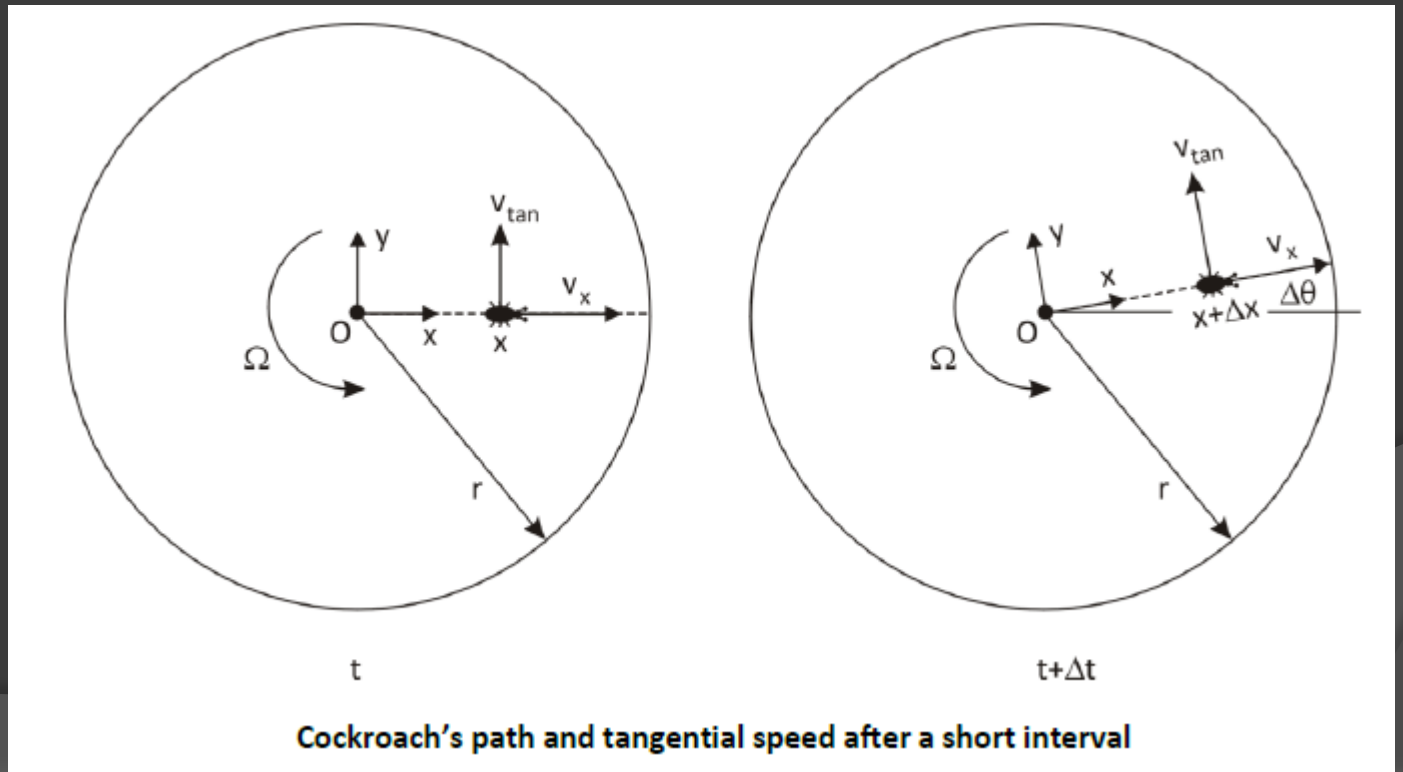
- A cockroach is moving on a old Vinyl LP record.



Cockroach walking out on a rotating LP platter

Coriolis Component of Acc.

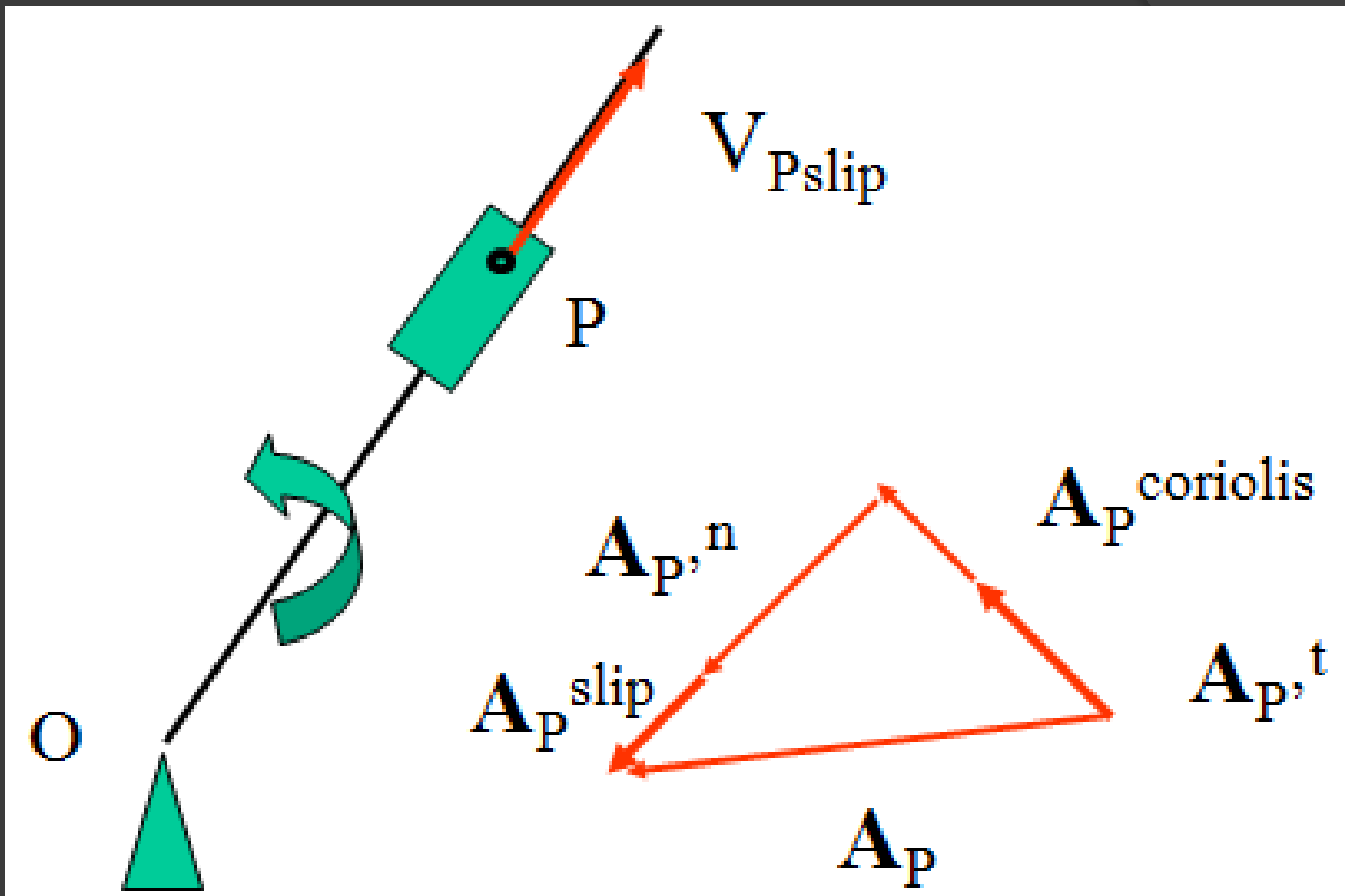
- Let's consider the simple situation where the rotational speed Ω is constant and the cockroach's walking velocity is also constant.
- The following two fig shows position of cockroach at instant t and t+Δt.



Cockroach's path and tangential speed after a short interval

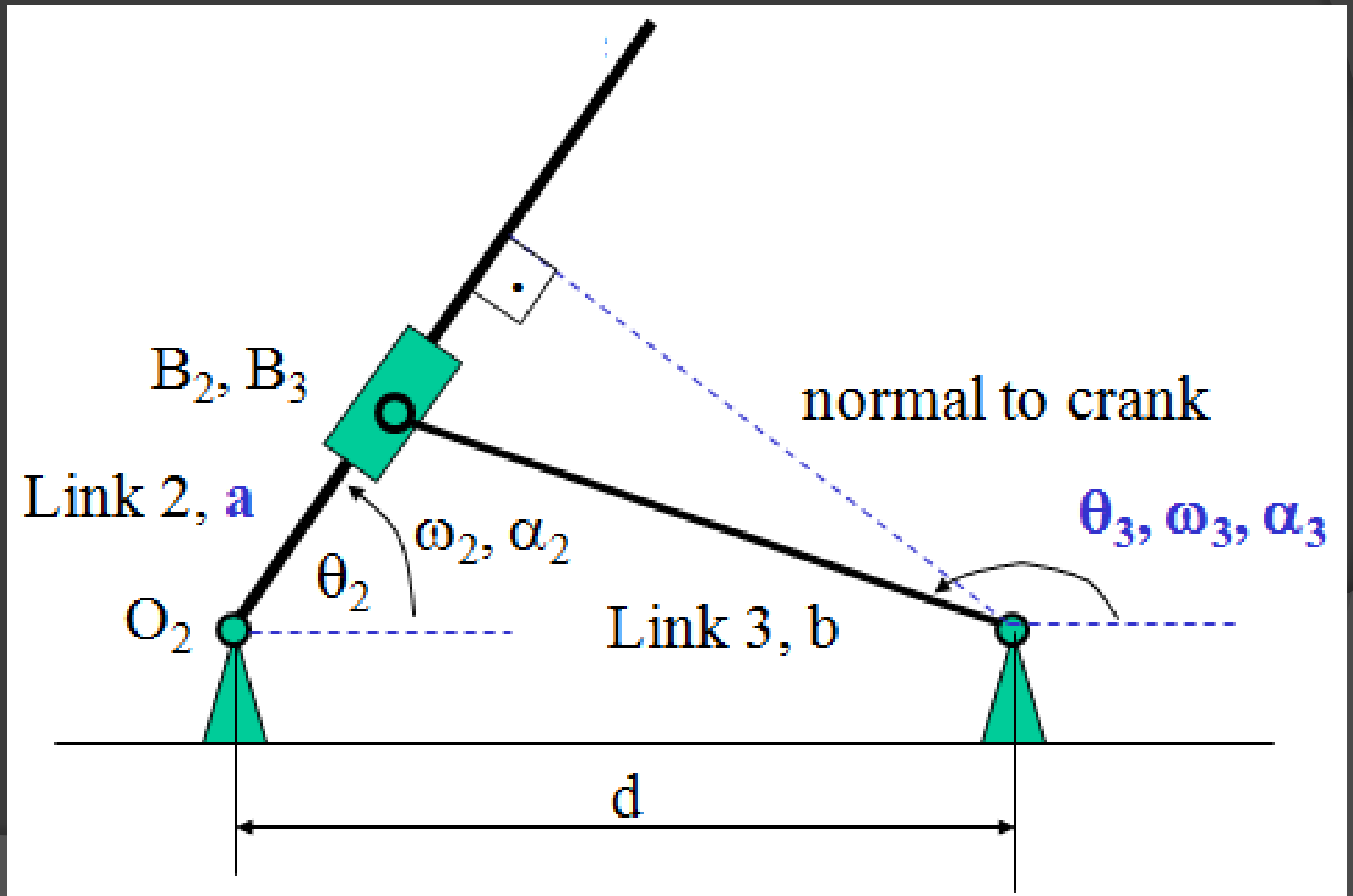
Coriolis Component of Acc.

- Coriolis acceleration = $2 \cdot \omega \cdot V_{slip}$
- Coriolis acceleration is normal to the radius, OP, and it points towards the left of an observer moving with the slider if rotation is counterclockwise. If the rotation is clockwise it points to the right.
- To find the acceleration of a point, P, moving on a rotating path: Consider a point, P', that is fixed on the path and coincides with P at a particular instant. Find the acceleration of P', and add the slip acceleration of P and the Coriolis acceleration of P.
- $\mathbf{A}_P$=acceleration of P' + acceleration of P seen from observer moving with rod + Coriolis acceleration=$\mathbf{A}_{P'} + \mathbf{A}_p^{slip} + \mathbf{A}_P^{Coriolis}$


V_{Pslip}
P
O
$A_P^{coriolis}$
$A_{P,}^{n}$
A_P^{slip}
$A_{P,}^{t}$
A_P

Coriolis Component of Acc.

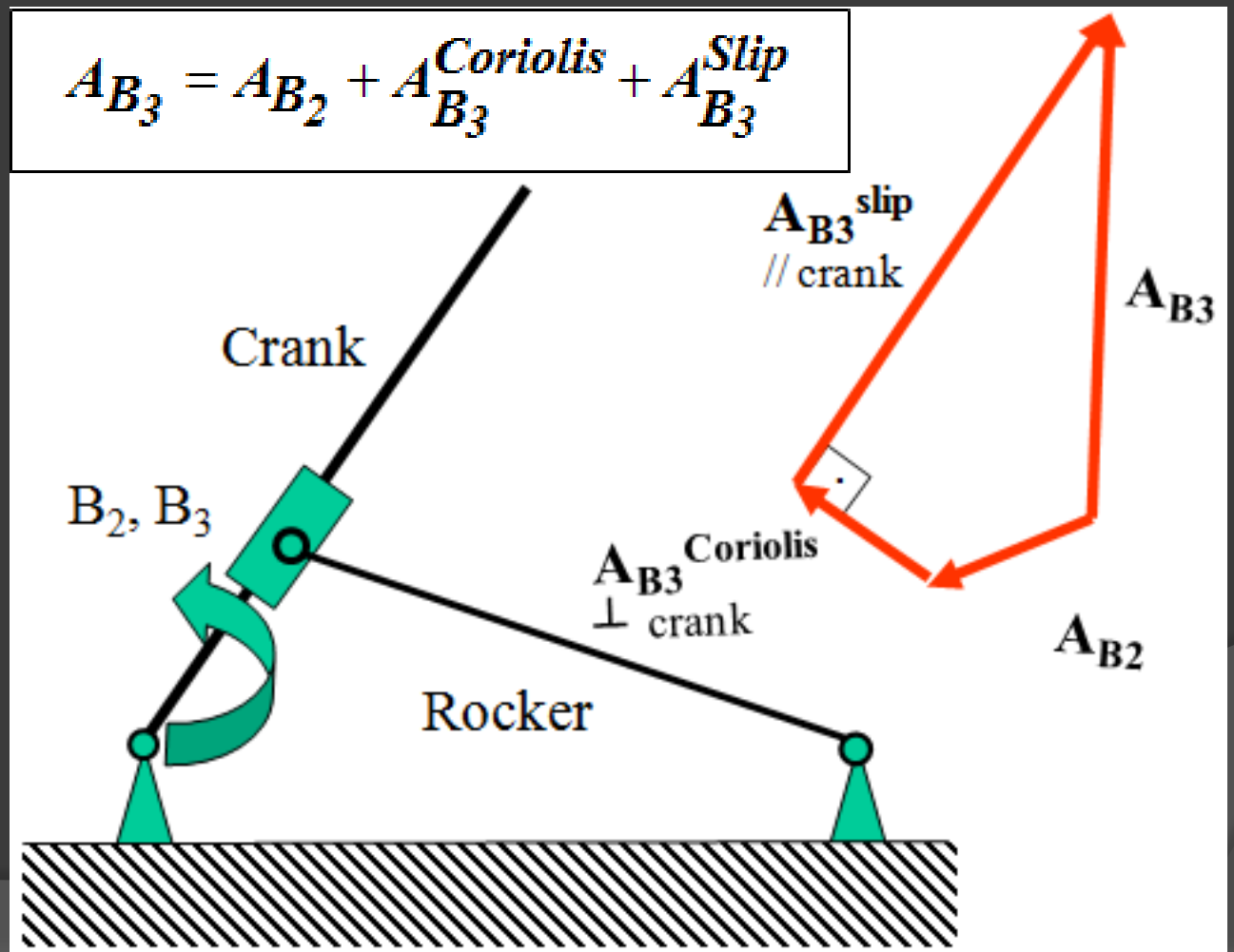
- Application : Slotted Lever mechanism.



Coriolis Component of Acc.

- Relation between accelerations of B_2 (on crank) and B_3 (on slider)

$$A_{B_3} = A_{B_2} + A_{B_3}^{Coriolis} + A_{B_3}^{Slip}$$



Thank You

